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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Civil) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

20CVE703/17CV613 - RURAL WATER SUPPLY AND SANITATION

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	i) Infer the advantages of protected water supply. ii) Explain major causes of groundwater pollution.	10	L*2	1	1
b)	Explain Eco-sanitation highlighting the advantages.	10	L2	2	1
2. a)	Explain adsorption technique of fluoride removal Outlining advantages of Nalgonda technique of fluoride removal.	10	L2	1	1
b)	Sketch and explain the concept of wastewater treatment in septic tank.	10	L2	2	2
3. a)	Sketch and explain zeolite process of water softening.	10	L2	1	1
b)	With the help of neat diagram explain Two pit system of sanitation applicable to rural area.	10	L2	2	1
Unit – II					
4. a)	Sketch and explain the life cycle of housefly.	10	L2	3	1
b)	Explain the following solid waste collection services. i) Curb services ii) Alley services iii) Set out - Set back services iv) Set out services v) Backyard services.	10	L2	4	1
5. a)	Define the following terms. Infection, Contamination, Infestation, Contagious disease, Host, Vector of inspection, Epidemic, Endemic, Eradication, Elimination.	10	L1	3	1
b)	Examine the strategies for disposing of solid waste listed below. i) Bangalore method of composting ii) Indoré method of composting.	10	L4	4	1
6. a)	Explain chain of infection with neat sketch and discuss the methods of communication of diseases.	10	L2	3	1
b)	Sketch and explain mass burning system of incineration.	10	L2	4	1
Unit – III					
7. a)	Examine the factors to be considered when building a cowshed.	10	L4	5	2
b)	Explain the following cattle diseases. i) Anthrax ii) Brucellosis	10	L2	5	1
8. a)	Explain the following tests on milk. i) Alcohol test ii) Lactometer test	10	L2	5	1
b)	i) Explain the significance of pasteurization. ii) Explain the clarification stage and Homogenization stages of milk pasteurization.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
